

Ambr 250 Modular - microbial dual-Rushton vessel, 100-250 mL

sartorius-ambr-250

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	100 mL	Source-supported
Maximum working volume	250 mL	Source-supported
Nominal vessel / bag size	250 mL	Source-supported
Total vessel volume loaded into the form	250 mL	Derived / estimated
Initial working volume loaded into the form	100 mL	Derived / estimated
Product-contact material	Plastic / single-use	App assumption
Height : diameter	3	App assumption
Internal / chamber diameter	0.0473 m	Derived / estimated
Liquid height at maximum working volume	0.142 m	Derived / estimated
Impeller / mixing type	Six-blade Rushton turbine	Source-supported
Impeller count	2	Source-supported
Impeller diameter	0.0156 m	Derived / estimated
Power number	5	App assumption
Initial agitation	1000 rpm	App assumption
Maximum agitation	4500 rpm	App assumption
Initial aeration	0.5 vvm	App assumption
Maximum aeration	2 vvm	App assumption
Baffled vessel	Yes	App assumption
Maximum pressure	0.5 bar(g)	App assumption
Default operating pressure	0.1 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 100 mL

Source-supported | Sartorius Ambr 250 Modular brochure - PDF page 7; Single-use mammalian or microbial bioreactors

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 100 mL.

Limitations: Selected microbial dual-Rushton configuration; not the single-impeller unbaffled variant. No model-specific speed limit verified.

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[Manufacturer publication](#)

Maximum working volume - 250 mL

Source-supported | Sartorius Ambr 250 Modular brochure - PDF page 7; Single-use mammalian or microbial bioreactors

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 250 mL.

Limitations: Selected microbial dual-Rushton configuration; not the single-impeller unbaffled variant. No model-specific speed limit verified.

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Nominal vessel / bag size - 250 mL

Source-supported | Sartorius Ambr 250 Modular brochure - PDF page 7; Single-use mammalian or microbial bioreactors

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 250 mL.

Limitations: Selected microbial dual-Rushton configuration; not the single-impeller unbaffled variant. No model-specific speed limit verified.

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Total vessel volume loaded into the form - 250 mL

Derived / estimated | Form-population rule

Rule: total volume = max(nominal vessel volume, maximum working volume)

Selection / calculation: $\max(250 \text{ mL}, 250 \text{ mL}) = 250 \text{ mL}$

Limitations: This prevents the form's total vessel volume from being smaller than the working-volume ceiling.

Initial working volume loaded into the form - 100 mL

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: $\min(100 \text{ mL}, 250 \text{ mL}) = 100 \text{ mL}$

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - Plastic / single-use

App assumption | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: Plastic / single-use.

Limitations: No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Height : diameter - 3

App assumption | Generic microbial stirred-tank profile

Rule: PROFILE_DEFAULTS["microbial"].heightDiameterRatio = 3

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microbial stirred-tank starting profile. This ratio is subsequently used to estimate vessel diameter and liquid height.

Internal / chamber diameter - 0.0473 m

Derived / estimated | Cylindrical geometry estimate

Rule: $D = \sqrt[3]{(4 \times V) / (\pi \times H:D)}$

Selection / calculation: $D = \sqrt[3]{(4 \times 0.00025 \text{ m}^3) / (\pi \times 3)} = 0.0473416 \text{ m}$; the form displays 0.0473 m.

Limitations: The estimate treats the liquid volume as a straight-sided cylinder and does not include heads, cones, internals or manufacturer-specific dimensions.

Liquid height at maximum working volume - 0.142 m

Derived / estimated | Cylindrical geometry estimate

Rule: liquid height = estimated internal diameter x H:D

Selection / calculation: $0.0473416 \text{ m} \times 3 = 0.1420248 \text{ m}$; the form displays 0.142 m.

Limitations: This is linked to the same ideal-cylinder approximation used for the diameter.

Impeller / mixing type - Six-blade Rushton turbine

Source-supported | Sartorius Ambr 250 Modular brochure - PDF page 7; Single-use mammalian or microbial bioreactors

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Six-blade Rushton turbine.

Limitations: Selected microbial dual-Rushton configuration; not the single-impeller unbaffled variant. No model-specific speed limit verified.

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Impeller count - 2

Source-supported | Sartorius Ambr 250 Modular brochure - PDF page 7; Single-use mammalian or microbial bioreactors

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 2.

Limitations: Selected microbial dual-Rushton configuration; not the single-impeller unbaffled variant. No model-specific speed limit verified.

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Impeller diameter - 0.0156 m

Derived / estimated | Generic impeller-to-tank ratio

Rule: impeller diameter = $\max(0.33 \times \text{estimated vessel diameter}, 0.001 \text{ m})$

Selection / calculation: $\max(0.33 \times 0.0473416 \text{ m}, 0.001 \text{ m}) = 0.01562273 \text{ m}$; the form displays 0.0156 m.

Limitations: A one-third tank-diameter impeller is used only as a generic starting geometry when no model-specific impeller diameter is stored.

Power number - 5

App assumption | Generic microbial stirred-tank profile

Rule: PROFILE_DEFAULTS["microbial"].powerNumber = 5

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microbial stirred-tank starting profile. The power number is an empirical generic value used in $P = N_p N^3 D_i^5$. It is not recalculated from blade geometry.

Initial agitation - 1000 rpm

App assumption | Vessel-specific preset override

Rule: `baseRpm = 1000`

Selection / calculation: No calculation. The value is written directly into this vessel's catalogue entry.

Limitations: This vessel-specific value overrides the generic microbial stirred-tank profile. It is still treated as an app assumption unless the parameter is explicitly listed as source-supported. This is the initial agitation setting loaded into the form.

Maximum agitation - 4500 rpm

App assumption | Vessel-specific preset override

Rule: `maxRpm = 4500`

Selection / calculation: No calculation. The value is written directly into this vessel's catalogue entry.

Limitations: This vessel-specific value overrides the generic microbial stirred-tank profile. It is still treated as an app assumption unless the parameter is explicitly listed as source-supported. This is the ceiling available to the generic DO-control cascade.

Initial aeration - 0.5 vvm

App assumption | Generic microbial stirred-tank profile

Rule: `PROFILE_DEFAULTS["microbial"].baseVvm = 0.5`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microbial stirred-tank starting profile. This is the initial gas rate loaded into the form.

Maximum aeration - 2 vvm

App assumption | Generic microbial stirred-tank profile

Rule: `PROFILE_DEFAULTS["microbial"].maxVvm = 2`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microbial stirred-tank starting profile. This is the gas-rate ceiling available to the generic DO-control cascade.

Baffled vessel - Yes

App assumption | Generic microbial stirred-tank profile

Rule: `PROFILE_DEFAULTS["microbial"].baffled = true`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microbial stirred-tank starting profile. This Boolean changes the generic vessel configuration and mixing interpretation.

Maximum pressure - 0.5 bar(g)

App assumption | Generic microbial stirred-tank profile

Rule: `PROFILE_DEFAULTS["microbial"].maxPressure = 0.5`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microbial stirred-tank starting profile. This is the pressure ceiling used for warnings in the generic equipment model.

Default operating pressure - 0.1 bar(g)

App assumption | Generic microbial stirred-tank profile

Rule: `PROFILE_DEFAULTS["microbial"].operatingPressure = 0.1`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microbial stirred-tank starting profile. This is the starting headspace pressure loaded into the form.

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: maxOxygenFraction = 100

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

The two-impeller microbial configuration and high agitation ceiling are source-supported; the selected starting RPM remains a simulator starting point.

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

Reviewed De Wilde et al., Superior Scalability of Single-Use Bioreactors, BPI September 2014, PDF pp. 1-6 (printed 14-19).

Manufacturer-authored characterization article, not a current hardware datasheet. Archived source:

sartorius-dewilde-scalability-2014-27b44451dc52.pdf

Historical ambr250 cell-culture evidence is not matched to this microbial dual-Rushton Modular preset. Table 1 reports 60-250 mL working, 360 mL total, vessel D 62.5 mm, liquid height 90 mm and impeller D 26 mm. These values are NOT applied. The article draws ambr characterization from Bareither et al. 2013, reference 20; 200-800 rpm are study conditions, not hardware ratings.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Supplementary archived source: Superior Scalability of Single-Use Bioreactors - De Wilde et al., September 2014 - PDF page 2

Historical comparison only; not blanket validation of current model

Local original document identities

Sartorius Ambr 250 Modular brochure

sartorius-ambr-250-modular-brochure-2e7a221f0fc7.pdf

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Superior Scalability of Single-Use Bioreactors - De Wilde et al., September 2014

sartorius-dewilde-scalability-2014-27b44451dc52.pdf

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